

Major Project: Seasonal Agriculture Performance Analysis

Objective

To analyze agricultural data from different seasons and identify meaningful patterns, trends, relationships and differences in agricultural performance.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

# Set visualization style
sns.set_theme(style="whitegrid", palette="muted")
%matplotlib inline

# Load the dataset
df = pd.read_csv('seasonal_agriculture_performance_dataset.csv')
print(f"Dataset Shape: {df.shape}")
display(df.head())
```

Dataset Shape: (4000, 28)

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_hrs
0	SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif	0.53	486.4	24.3	69.1	12.5
1	SF10002	Maharashtra	Nalgonda	Maize	Kharif	6.53	855.4	25.7	78.4	11.2
2	SF10003	Telangana	Warangal	Pulses	Rabi	4.86	455.6	23.9	56.4	13.8
3	SF10004	Telangana	Indore	Rice	Kharif	4.50	753.2	29.7	65.2	10.1
4	SF10005	Karnataka	Ludhiana	Maize	Zaid	4.21	101.6	34.1	55.2	9.7

5 rows × 28 columns

1. Data Cleaning and Preparation

We will check for missing values and understand the data types.

```
# Check data types and missing values
df.info()

# Basic statistical summary
display(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4000 entries, 0 to 3999
Data columns (total 28 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Farm_ID                               4000 non-null   object
1   State                                 4000 non-null   object
2   District                             4000 non-null   object
3   Crop                                  4000 non-null   object
4   Season                                4000 non-null   object
5   Farm_Area_Hectares                    4000 non-null   float64
6   Rainfall_mm                           3952 non-null   float64
7   Avg_Temperature_C                     4000 non-null   float64
8   Humidity_pct                          4000 non-null   float64
9   Sunlight_Hours_Day                   4000 non-null   float64
10  Soil_pH                               4000 non-null   float64
11  Soil_Moisture_pct                     3960 non-null   float64
12  Nitrogen_kg_ha                       4000 non-null   float64
13  Phosphorus_kg_ha                     4000 non-null   float64
14  Potassium_kg_ha                      4000 non-null   float64
15  Irrigation_Method                     4000 non-null   object
16  Fertilizer_kg_ha                     4000 non-null   float64
17  Pesticide_Litre_ha                   4000 non-null   float64
18  Seed_Quality_Score                   4000 non-null   float64
19  Yield_Tonnes_Ha                      3968 non-null   float64
20  Production_Tonnes                    4000 non-null   float64
21  Market_Price_INR_Tonne               4000 non-null   int64
22  Total_Cost_INR                       4000 non-null   int64
23  Revenue_INR                          4000 non-null   int64
24  Profit_INR                           4000 non-null   int64
25  Water_Used_m3                        4000 non-null   int64
26  Water_Efficiency_t_per_1000m3        4000 non-null   float64
27  Disease_Pest_Risk_pct                 4000 non-null   float64
dtypes: float64(17), int64(5), object(6)
memory usage: 875.1+ KB
```

	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	Soil_pH	Soil_Moisture_pct
count	4000.000000	3952.000000	4000.000000	4000.000000	4000.000000	4000.000000	3960.000000
mean	7.888265	601.152657	26.81890	63.210500	7.323050	6.701010	26.508232
std	4.223764	288.925025	3.81818	11.960834	1.121186	0.641461	6.712484
min	0.500000	80.000000	16.20000	25.000000	3.500000	5.200000	8.000000
25%	4.260000	370.200000	23.90000	54.775000	6.600000	6.260000	21.700000
50%	7.895000	582.000000	26.90000	63.000000	7.300000	6.690000	26.400000
75%	11.650000	834.950000	29.60000	71.900000	8.100000	7.140000	31.300000
max	15.000000	1395.200000	39.70000	95.000000	11.000000	8.400000	47.000000

8 rows × 22 columns

2. Exploratory Data Analysis (EDA) & Seasonal Variations

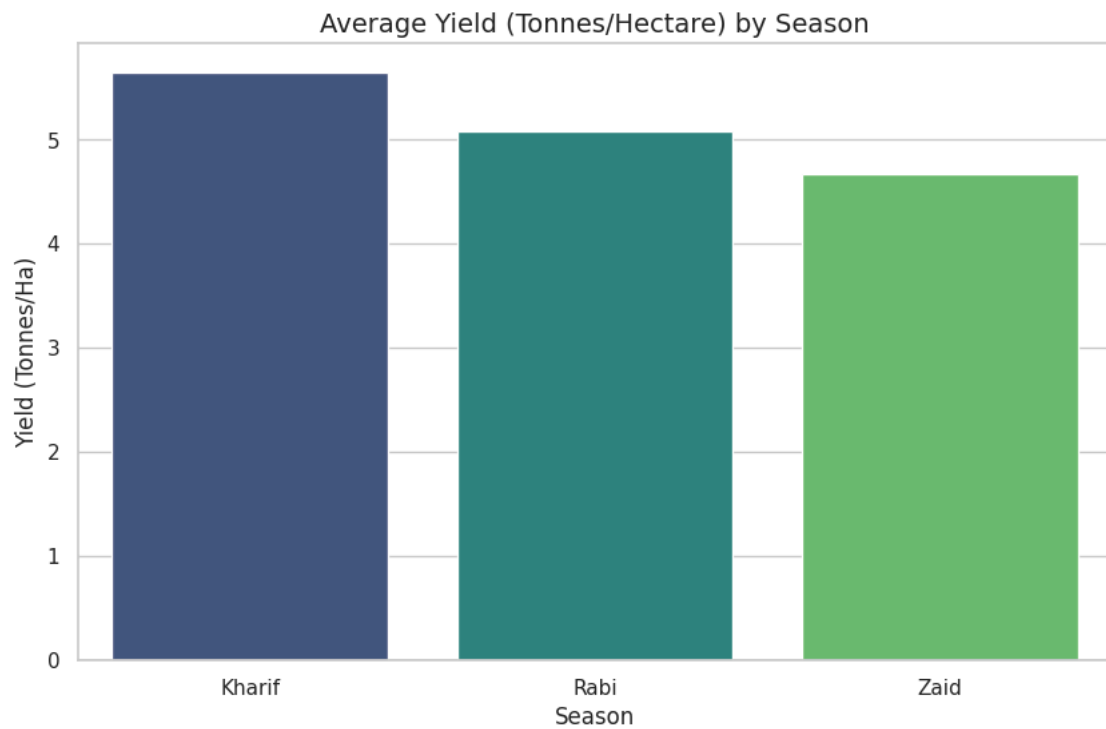
Let's analyze how agricultural performance (Yield, Profit, Revenue) varies across seasons.

```
# Grouping by Season to see the average performance metrics
seasonal_metrics = df.groupby('Season')[['Yield_Tonnes_Ha', 'Profit_INR', 'Revenue_INR', 'Total_Cost_INR']].mean().reset_index()
display(seasonal_metrics)
```

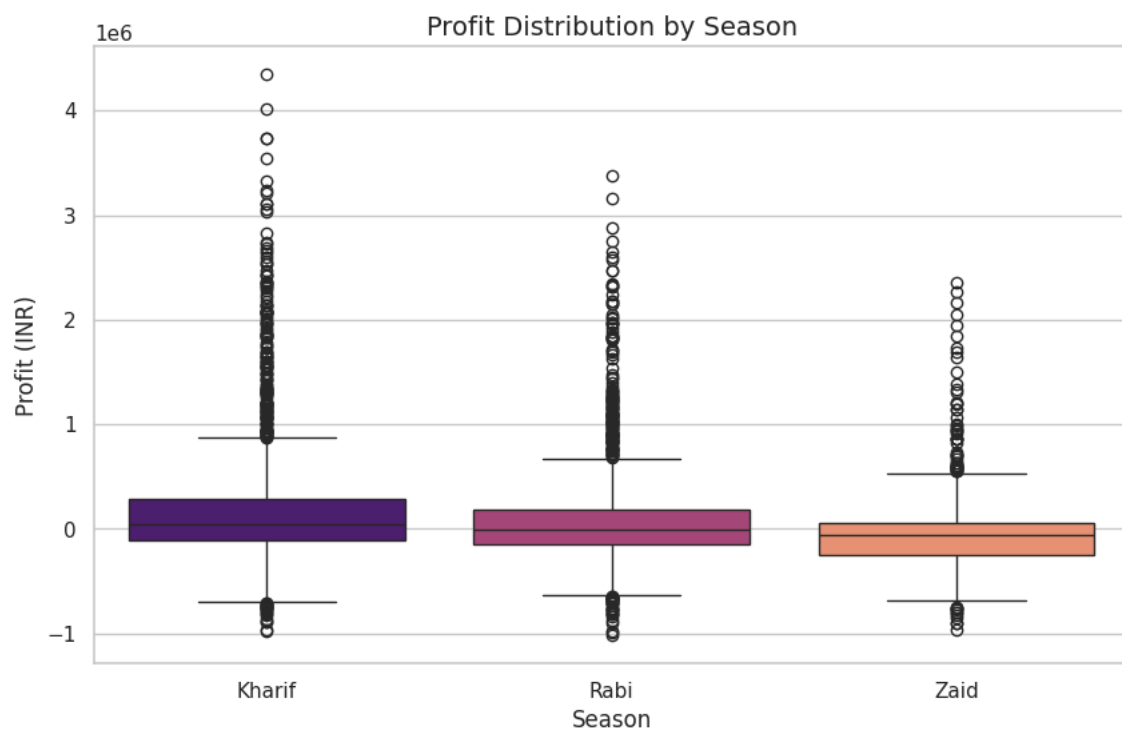
	Season	Yield_Tonnes_Ha	Profit_INR	Revenue_INR	Total_Cost_INR
0	Kharif	5.643973	178914.647555	710719.055649	531804.408094
1	Rabi	5.080498	87689.470191	601526.048556	513836.578365
2	Zaid	4.665942	-24804.824916	519171.904040	543976.728956

Next steps: [Generate code with seasonal_metrics](#) [New interactive sheet](#)

```
# Visualizing Yield across Seasons
plt.figure(figsize=(10, 6))
sns.barplot(data=df, x='Season', y='Yield_Tonnes_Ha', ci=None, palette='viridis')
plt.title('Average Yield (Tonnes/Hectare) by Season', fontsize=14)
plt.ylabel('Yield (Tonnes/Ha)')
plt.xlabel('Season')
plt.savefig('yield_by_season.png')
plt.show()
```



```
# Visualizing Profit across Seasons
plt.figure(figsize=(10, 6))
sns.boxplot(data=df, x='Season', y='Profit_INR', palette='magma')
plt.title('Profit Distribution by Season', fontsize=14)
plt.ylabel('Profit (INR)')
plt.xlabel('Season')
plt.savefig('profit_by_season.png')
plt.show()
```



3. Resource Usage Across Seasons

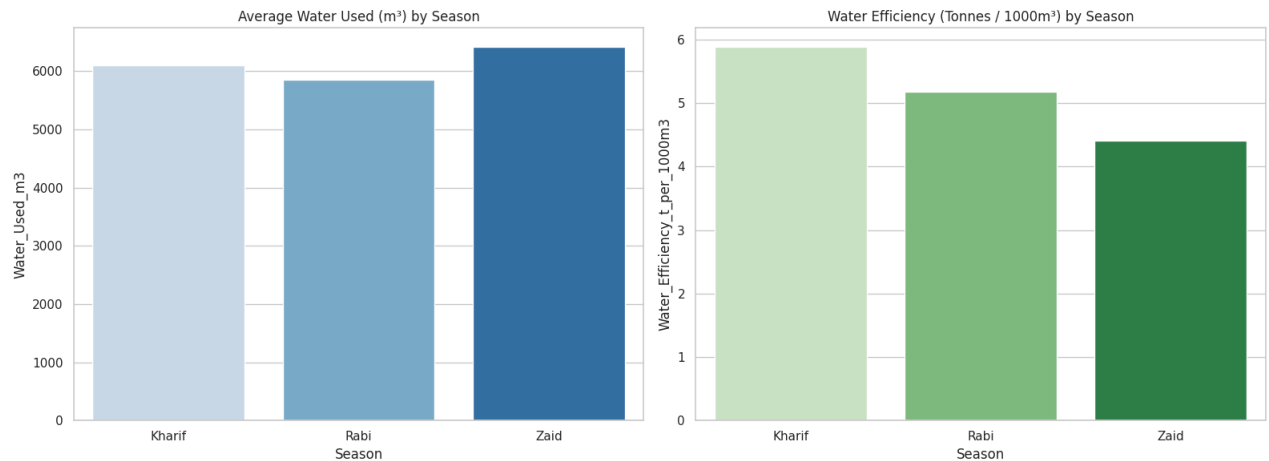
Are there noticeable variations in resource usage (Water, Fertilizers, Pesticides) across seasons?

```
# Visualizing Water Usage and Efficiency
fig, ax = plt.subplots(1, 2, figsize=(16, 6))

sns.barpplot(data=df, x='Season', y='Water_Used_m3', ci=None, palette='Blues', ax=ax[0])
ax[0].set_title('Average Water Used (m³) by Season')
```

```
sns.barplot(data=df, x='Season', y='Water_Efficiency_t_per_1000m3', ci=None, palette='Greens', ax=ax[1])
ax[1].set_title('Water Efficiency (Tonnes / 1000m³) by Season')
```

```
plt.tight_layout()
plt.savefig('resource_usage.png')
plt.show()
```



4. Environmental Conditions and Performance

Let's check the correlation between environmental factors and yield/profit.

```
# Select numerical columns for correlation
env_cols = ['Rainfall_mm', 'Avg_Temperature_C', 'Humidity_pct', 'Sunlight_Hours_Day', 'Soil_Moisture_pct', 'Yield_Tonnes_Ha']
corr_matrix = df[env_cols].corr()

plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=.5)
plt.title('Correlation Matrix: Environment vs Performance', fontsize=14)
plt.savefig('correlation_heatmap.png')
plt.show()
```

